

Tuesday Group 7

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CE331 Group Project: Mapping of the GI Lab And the ACES Building

Objective

The objective of this survey project was to create an accurate and detailed map of the GI lab and ACES building and its nearby area using Total Station along with auto-level to create a map (using QGIS software).

Procedure

We started by first exploring the area to understand the site and its surroundings. After that, we identified change points and performed leveling to establish accurate elevation reference points. This was followed by traversing to measure angles and distances, forming a strong base for mapping. Finally, detailed feature mapping was conducted to record important buildings and nearby features, helping us create a clear, accurate map of the area.

Methodology

Initial brief

On 7th October (Day 1) we were briefed about the group project. Our group was assigned the region surrounding the GI lab and ACES building.

We performed the following operations during our project :

1. Reconnaissance
2. Setup CPs
3. Traversing
4. Levelling
5. Feature Mapping

After this, we transferred the .csv file carrying the survey data to a pendrive and used it to create the map using the QGIS software.

Week 1

We carried out reconnaissance of the region surrounding the GI lab and ACES building and decided on 5 Control Points that were inter-visible and relatively distant. We took care to select two CPs open to the sky and clear of obstructions to be able to obtain GPS coordinates for them next week. We marked these CPs with orange paint, encircled them and labelled them “TE7-#” (where # is the control point number).



Figure 1: Image of the area with all control points (source: Google Maps)

Reconnaissance was the initial inspection or preliminary survey of the area to be surveyed. We did a systematic assessment of the site's physical, environmental, and logistical aspects to plan and organize the survey effectively.

This helped us in setting up control points at the most suitable locations, considering features in the area, inter-visibility and points being open to sky

We finalized 5 control points and then marked them using Orange paint.

Week 2

In week 2, we carried out levelling operations to calculate the elevations of our control points.

We started from a benchmark at an elevation of 128.408m and made a closed traversal back to the benchmark in order to obtain the misclosure error.

Levelling operations involved the use of an auto-level and a measuring staff (along with tripod). Backsight and foresight readings were taken and distances were calculated through the US and LS readings.

Methodology:

1. Level the tripod of the Auto-level instrument manually.
2. Place the Auto level on the tripod. And level it using the foot screws.
3. Take the first reading (back sight) by putting staff on the known RL (near GI Lab) as this is the benchmark.
4. Now move the staff to the first Control point such that the distance between previous staff location and Auto level instrument is same as the current position of staff and Auto level instrument. Note down the reading on the field book.
5. Now place the Auto level instrument on the second control point. And take the reading without disturbing the previous location of the staff.
6. Similarly, carry out these steps to complete the whole loop getting the reduced level of every control point.



Figure 2: [Auto-level](#), [Tripod](#) and [Leveling Staff](#) (source-indiamart.com)

CALCULATIONS (LEVELLING)

Tachometric Distance Calculation:

$$d_{\text{leveled}} = (US - LS) \times 100$$

Where:

$$d_{\text{leveled}} = \text{Distance leveled (in m)}$$

US = Upper stadia reading (in m)

LS = Lower stadia reading (in m)

K = 100 = stadia constant

Error Distribution (Adjustment) :

$$c_i = M \times i / \text{no. of Change Points}$$

Where:

c_i = error correction for the i^{th} change point

M = Misclosure

Adjusted Elevation at Change Point:

$$H_{i(c)} = H_i + c_i$$

Where:

$H_{i(c)}$ = Corrected Height

1. Level Adjustments

$$HI = RL + BS$$

$$= 128.409 + 1.078$$

$$= 129.487$$

$$RL = HI - FS$$

$$= 129.487 - 1.4$$

$$= 128.087$$

$$\text{DISTANCE} = (US - LS) \times 100$$

$$= (1.138 - 1.018) \times 100$$

$$= 12 \text{ m}$$

$$D = \sum D_i$$

$$= 12 + 18.1 + 13.8 + 17.1 + 25.5 + 21.4 + 33.2 + 16.5 + 5 + 12.1 + 21.8 + 14.1 + 23.9 + 24 + 21.9 + 29.1 = 339 \text{ m}$$

$$c[i] = i \times 0.006/9$$

$$= 0.0006666666667 \times i$$

$$\text{Adjusted Elevation} = RL + c[i]$$

$$= 128.087 + 0.0006666666667 \times 1$$

$$= 128.0876667$$

Quality of work (for levelling):

Total distance (k) in km = 339 m = 0.339 km

Tolerance (mm) = $c \times \sqrt{k} = 10.3050 \times \sqrt{(0.339)}$

c is around 10, hence quality is **Accurate**.

Misclosure = Sigma BS - Sigma FS = -6 mm

BS	FS	RISE	FALL	RL	REMARK	US(BS)	LS(BS)	US(FS)	LS(FS)	Distance (m)			Cumulative Distance (m)	Correction(m)	Adjusted RL(m)
										D1	D2	Total			
BM	1.078			128.409	BM	1.138	1.018			12			0	0	128.409
A	1.468	1.4	0.322	128.087		1.558	1.37	1.433	1.312	18.8	12.1	24.1	24.1	0.0006666666667	128.0876667
CP-4	1.3	1.302	0.166	128.253	CP4	1.369	1.231	1.413	1.195	13.8	21.8	40.6	64.7	0.001333333333	128.2543333
CP-3	1.431	1.342	0.042	128.211	CP3	1.519	1.348	1.414	1.273	17.1	14.1	27.9	92.6	0.002	128.213
CP-2	1.333	1.482	0.051	128.16	CP2	1.465	1.21	1.604	1.365	25.5	23.9	41	133.6	0.002666666667	128.1626667
B	1.312	1.445	0.112	128.048		1.419	1.205	1.565	1.325	21.4	24	49.5	183.1	0.003333333333	128.0513333
CP-1	1.325	1.315	0.003	128.045	CP1	1.491	1.159	1.424	1.205	33.2	21.9	43.3	226.4	0.004	128.049
CP-5	1.384	1.404	0.079	127.966	CP5	1.465	1.3	1.55	1.259	16.5	29.1	62.3	288.7	0.004666666667	127.9706667
D	1.37	1.287	0.097	128.063		1.405	1.355	1.372	1.203	5	16.9	33.4	322.1	0.005333333334	128.0683333
BM		1.03	0.34	128.403	BM			1.089	0.97		11.9	16.9	339	0.006	128.409
SUM	12.001	12.007	0.603	0.609									339		
LHS	= $\sum BS - \sum FS$ =		-0.006	error(m)	-0.006	Misclosure= 6mm		Quality of work		c					
RHS	= $\sum Rise - \sum Fall$ =		-0.006	correction(m)	0.006			ordinary		100					
RHS	= Last RL - First RL =		-0.006	k(km)	0.339			Rough		24					
				$E=c(k)^{1/2}$	6			Accurate		12					
				c	10.3050808			Precise level		4					
				Quality: Accurate!				High Quality		1					

Week 3&4

We traversed the circuit formed by our CPs measuring the internal angles and horizontal distances of the thus formed polygon. We did this using a total station (on a tripod) and reflector prism (on a bipod).

Methodology:

1. Choose a control point among the 5 points and set up the total station at that point ensuring proper leveling and centering.
2. Perform the necessary operations on the total station to make it operational.
3. First we stationed at point 1(from figure), aligned the theodolite to sight the reflector at point 5 and set the horizontal angle to zero.
4. Next, rotate the theodolite to align with the reflector at point 2 and record the horizontal angle.
5. Rotate the theodolite back to point 5 in the same direction as before and take the reading.
6. Record both the horizontal angle and the horizontal distances between the points of the polygon.
7. Repeat the steps at each point of the polygon note down the reading.



Figure 3: [Total Station](#), [prism pole with surveying Bipod](#), [Reflector Prism](#) (source-attached)

CP	Station Observed	Face (L/R)	Readings	Distance	Horizontal Angle		Correction (")	Corrected Internal Angle
					Angle In Decimal Format	Internal Angle(°)		
1	1		0° 0' 0"	89.184	0			
	3		81° 50' 21"	41.328	81.83916667			
	1'		0° 0' 1"	89.184	0.0002777777778			
	Error	R	0° 0' 1"		0.0002777777778	81.83910352	0.00505585241	81.84415937
	2		0° 0' 00"	41.328	0			
2	2		186° 20' 15"	28.391	186.3375			
	4		359° 59' 57"	41.328	359.9991667			
	2'		0° 0' 3"		0.0008333333333	186.3379313	0.00505585241	186.3429872
	Error	R	0° 0' 0"	28.391	0			
	3		87° 19' 22"	93.11	87.32277778			
3	3		359° 59' 37"	28.391	359.9936111			
	Error	R	0° 0' 23"		0.006388888889	87.32432749	0.00505585241	87.32938334
	4		0° 0' 0"	93.11	0			
	1		84° 55' 25"	62.917	84.92361111			
	4'		359° 59' 48"	93.11	359.9966667			
4	4		0° 0' 12"		0.003333333333	84.92439744	0.00505585241	84.92945329
	Error	R	0° 0' 0"	62.917	0			
	5		99° 33' 17"	99.5472222				
	2		0° 0' 0"	62.917	0.02083333333			
	5'		0° 01' 15"		0.02083333333	99.54896095	0.00505585241	99.55401681
5	1							
	Error	R						
					Sum of all internal angles ----> (Uncorrected)	539.9747207	0.02527926205	540
					Actual Sum of Angles ----> {(n-2)*180 }	540		
					Angle Misclosure (in degree) ---->	-0.02527926205		
Quality of work					Angle Misclosure (in seconds) ---->	91.00534336		
	First Order	6	13.41640786					
	Second Order	15	33.54101966					
Third Order		30	67.08203932					

1. Observations

At CP2, the readings were taken towards:

- Station 1 and Station 3

2. Sum of All Internal Angles

For a traverse with n sides:

$$\text{Theoretical Sum of Internal Angles} = (n-2) \times 180^\circ$$

If $n = 5$,

$$=(5-2) \times 180 = 540^\circ$$

From the table,

Observed (Uncorrected) Sum = 539.9747207°

3. Angular Misclosure

$$\text{Angular Misclosure} = \text{Theoretical Sum} - \text{Observed Sum}$$

$$540^\circ - 539.9747207^\circ = 0.0252793^\circ$$

(in seconds: $0.0252793 \times 3600 = 91.00534336$)

4. Correction per Station

Correction distributed equally to each angle:

$$\text{Correction per Station} = \text{Total Misclosure} / n$$

$$0.0252793^\circ / 5 = 0.00505585^\circ$$

5. Corrected Internal Angle

$$\text{Corrected Angle} = \text{Observed Internal Angle} + \text{Correction}$$

$$81.83910352^\circ + 0.00505585^\circ = 81.84415937^\circ$$

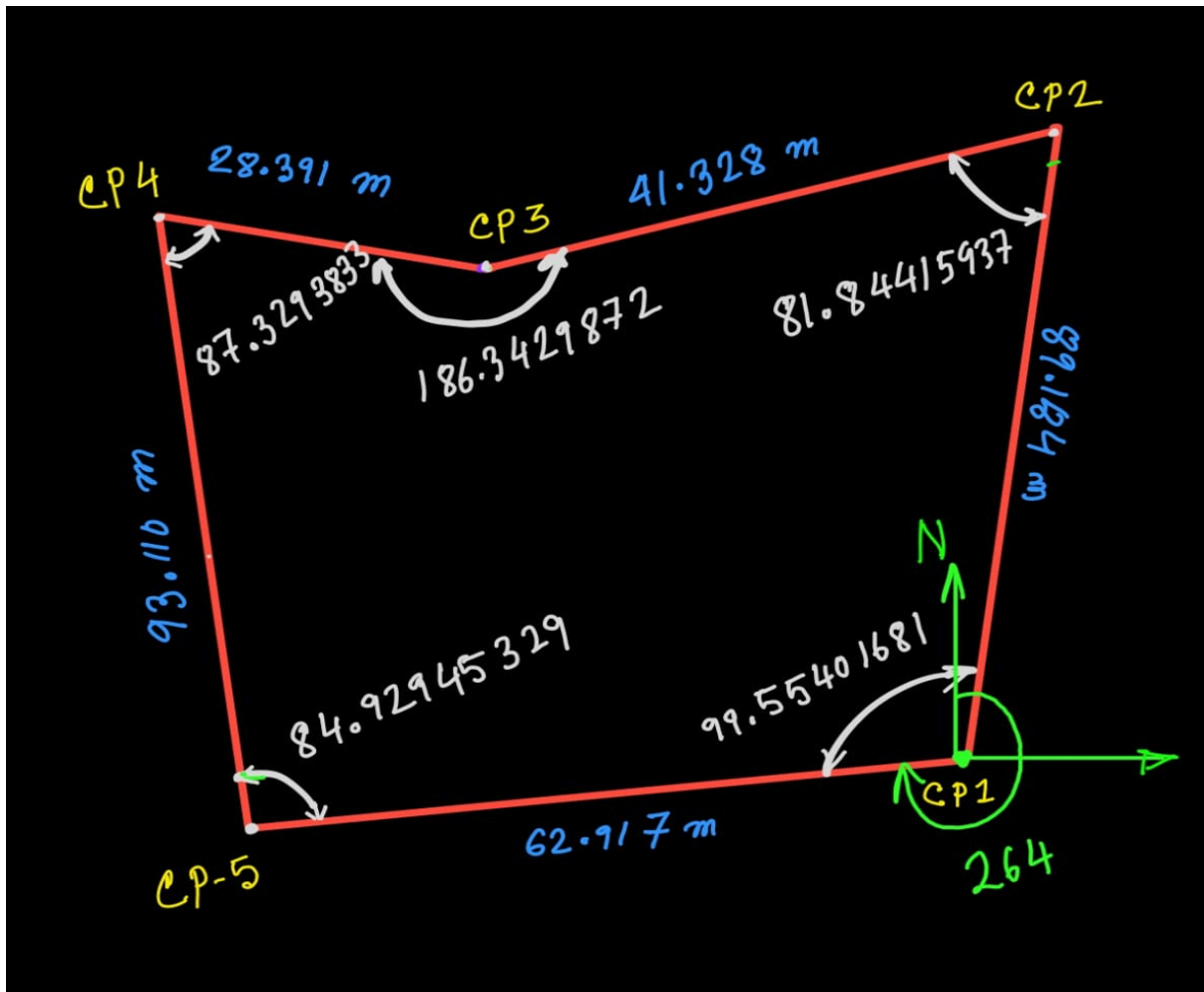


Figure 4: On field recorded data

Line	WCB (in degrees)	Internal Angle (in degree)	Length(m)	Latitude			Departure		
				Calculated	Correction	Corrected	Calculated	Correction	Corrected
1-2	183.5540168	99.55401681	89.184	5.517541	0.010933	5.528474	89.02134547	-0.00899437	89.0123511
2-3	265.3881762	81.84415937	41.328	-41.199838	0.005066	-41.194772	-3.31166419	-0.00416804847	-3.315832195
3-4	271.7411634	186.3429872	28.391	-28.3813716	0.00348	-28.3778916	0.8659225692	-0.002863284592	0.8630592846
4-5	179.0705467	87.32938334	93.11	1.4990484	0.011415	1.5104634	-93.09449358	-0.00939031	-93.10388389
5-1	264	84.92945329	62.917	62.5260131	0.0077131	62.5337262	6.55065099	-0.0063452952	6.544305695
sum-->			314.93	0.0386071	0.0386071	0	-0.03176127	-0.03176127	0
0									
Relative Precision				linear misclosure					
				0.04999286391					
Quality of Traverse				theta(in rad)					
				-0.878525542					
				theta(in deg)					
				-38.76206402					
Quality of work									
First order				1:25000					
Second order				1:10000					
Third order				1:5000					

Latitude and Departure Calculation

1. Latitude and Departure Calculation

$$\text{Latitude} = L \times \cos(\text{WCB})$$

$$\text{Departure} = L \times \sin(\text{WCB})$$

For line 1–2:

$$\text{Latitude} = 89.184 \times \cos(183.5540168^\circ) = -89.01248134$$

$$\text{Departure} = 89.184 \times \sin(183.5540168^\circ) = -5.528474$$

These are shown in the Calculated columns.

2. Check Misclosure

$$\Sigma L = +0.0386071 ; \Sigma D = -0.03176127$$

Ideally, both should be zero for a perfectly closed traverse. Non-zero values indicate coordinate misclosure.

3. Linear Misclosure

$$\begin{aligned} \text{Linear Misclosure} &= \sqrt{(\Sigma L)^2 + (\Sigma D)^2} \\ &= \sqrt{(0.0386071)^2 + (-0.03176127)^2} = 0.04999286391 \end{aligned}$$

4. Relative Precision

$$\text{Relative Precision} = \text{Linear Misclosure} / \text{Perimeter}$$

Given Traverse Perimeter = 314.93 m,

$$\text{Relative Precision} = 0.0499929 / 314.93 = 1/6300$$

From the table:

$$\text{Quality of Traverse} = 1 : 6300$$

5. Correction by Bowditch's (Compass) Rule

Bowditch's Rule distributes misclosure proportionally to line lengths:

$$\text{Correction in Latitude for each line} = -\Sigma L \times L_i / \Sigma L_i$$

$$\text{Correction in Departure for each line} = -\Sigma D \times L_i / \Sigma L_i$$

For Line 1–2:

$$\text{Latitude Correction} = +0.0386071 \times (89.184 / 314.93) = 0.010933$$

$$\text{Departure Correction} = -(-0.03176127) \times 89.184 / 314.93 = -0.00899437$$

6. Corrected Latitude and Departure

$$\text{Corrected Latitude} = \text{Calculated Latitude} + \text{Correction}$$

$$\text{Corrected Departure} = \text{Calculated Departure} + \text{Correction}$$

So,

$$\text{Latitude}_{\text{corr}} = 5.517541 + 0.010933 = 5.528474$$

$$\text{Departure}_{\text{corr}} = 89.02134547 - 0.00899437 = 89.0123511$$

These are the final adjusted coordinates used for plotting the traverse closure.

7. Angular Misclosure

The misclosure direction is found using:

$$\theta = \tan^{-1}(\Sigma D / \Sigma L)$$

$$\theta = \tan^{-1}(-0.03176127 / 0.0386071) = -38.762^\circ$$

(in radians: -0.67626)

This gives the direction of misclosure from North.

Week 5&6

1. The survey observations collected using the Total Station were first exported and organized.
2. The collected point data (trees, poles, manholes, road edges, building corners, etc.) was imported into QGIS software.
3. Separate vector layers (point, line, and polygon) were created to represent different features on the map.
4. Each feature was assigned proper symbology and labels (e.g., buildings, water tanks, roads, trees, electric poles, manholes).
5. Contour lines were added to represent elevation differences across the site.
6. The Layout Manager was used to design the final map, including title block, north arrow, scale bar, IITK logo, legend, and group details.
7. Finally, the map was exported in high resolution for reporting and documentation.

Methodology for map making:

1. Collect readings for features such as Building Corners, Electric Poles, Trees, Road edges, etc. along with the terrain. Follow a proper naming convention to record features.
2. For example, Trees can be stored as T1, T2, T3, T4, and so on. This helps in the identification of different features while preparing a map.
Repeat the steps again from control point 2 and take the readings of the rest of the features.
Transfer the data from the total station to the computer in a .csv file.

Using QGIS software for map making:

Now convert the csv file into two files one for storing all points other than ground point and one store ground point.

1. import data for this firstly we will go to layer >> add layer>>Add delimited text layer.
Or simply press ctrl+L

2. then browse All points.csv file already saved then in geometry definition specify X field as Easting ,Y field as Northing ,Z field as elevation and M field as point number then set geometry CRS as EPSG:32644 - WGS 84 / UTM zone 44N.
3. Then click on add to add points on the sheet.
4. Repeat step 1 and 2 to add ground points.
5. now to label these points right click on that and select properties and select label>>single label then set the desired value then click on apply ,labels will be applied to points .
6. Now to add symbology right click on layers select properties and select symbology>>categorized choose the value as point code , symbol then click on apply>>ok
7. Create features using all point data go to layer>>create layer>>new Shape file layer>>save file name>>select desired geometry type>>ok
8. To add features click on toggle >>add a polygon/line/point. Then connect the points based on geometry type let say for polygon feature like stage, building connect all the points of stage then right click and give id let say 1 and name let say stage then click on ok to edit the appearance of feature go to layer>>properties>>symbology modify it as desired.
9. For contour again go to layer>>properties>>labels>>single label. Then set the value as elevation value then press ok.
10. To interpolate go to processing>>toolbox>>IDW interpolation>>parameters. In parameter select vector layer as ground points and interpolation attribute as elevation >>click on use Z coordinate for interpolation, leave the extent to default then save the interpolate then click on run>>close.
11. got o layers>>Interpolate>>properties>>symbology. Select render type as contour then choose contour interval as 0.1, index contour level as 1 and input down scaling as 4 then click on apply and then ok.
12. To convert that into a contour layer click on raster>>extraction>>contour>>parameters. In parameters select input as interpolated interval between contour line as 0.1 rest are left to default then choose location to save contours and save it. Then click on run then close the tab.
13. Now we can label these contours using step 8.
14. To print Map go to project>>new print layout >>enter the name and save it>>A blank page will be created.
15. Now add map, then add grid by enabling grid under item properties section and make necessary changes.
16. Add legend and finally print the map.

Using the above steps we drew the following map.

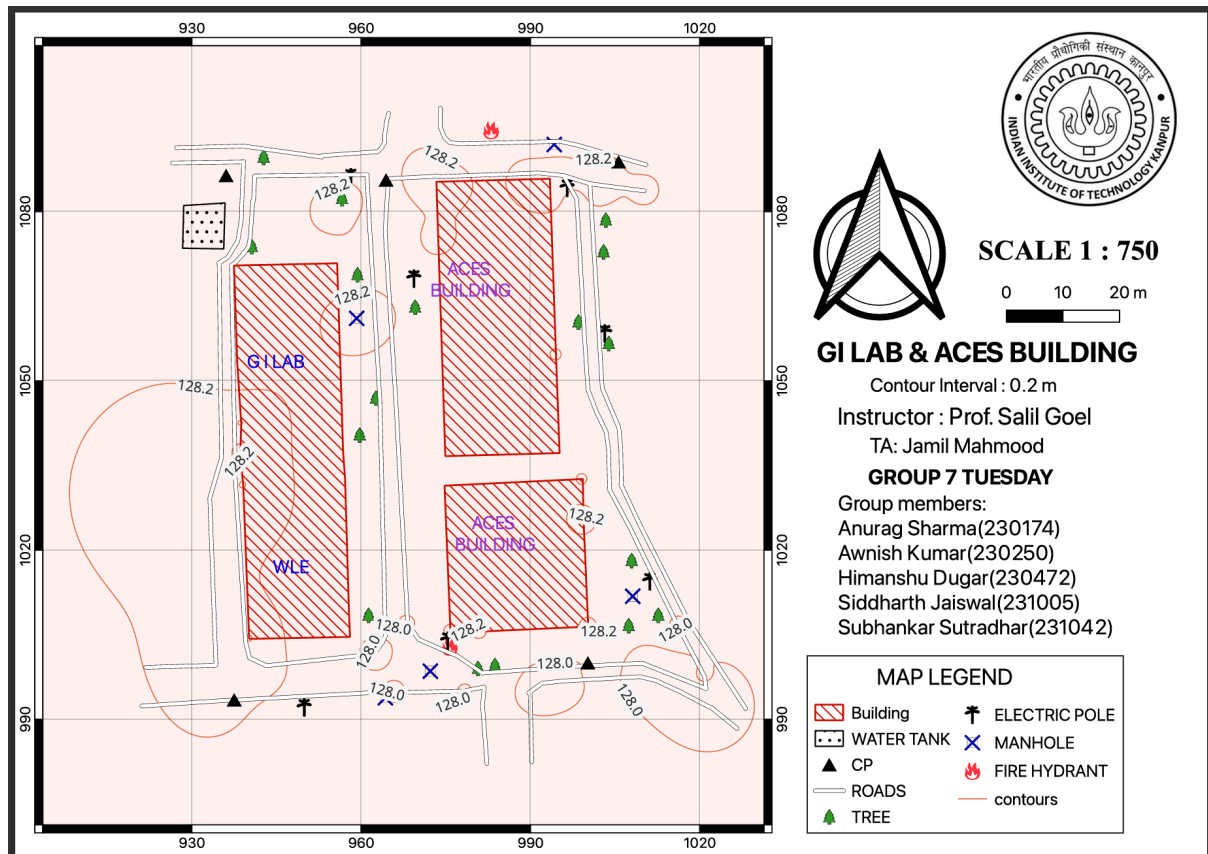


Figure 5: Map of GI lab and ACES building and different features around it.

Results and Discussion

1. We in our project successfully produced an accurate map of the ground and different features around the GI lab and ACES building.
2. Quality of work for levelling was accurate.
3. However, The traverse for angles was deemed Second Order in precision, while distance measurements was also of Second Order accuracy.

Conclusions

1. Making maps firsthand during this project taught us so much about what goes into the whole process.
2. We figured out the best possible ways to take readings and handled our findings using the right tools for the job.
3. For instance, we were facing a problem where a tree was coming in the line of sight of two CP's, then we discussed and made another CP between them and we were able to solve that problem.

4. To make sure everything was spot-on, we double-checked all our data and made sure to stay within the project's limits.
5. By the end of it all, we'd gotten much better at surveying and really sharpened our skills in checking data accuracy.
6. We can further improve the quality of our project by:
 - a. Picking more stable locations for control points, like concrete structures.
 - b. Keeping shorter distances between stations.
 - c. Keeping nearly equal backsight and foresight distances as much as possible.
 - d. Taking extra points around curved features

References

All the references used in preparing this lab report are as follows:

1. maps.google.com
2. QGIS software for map making.
3. Lecture notes by Professor Salil Goel.
4. https://www.indiamart.com/proddetail/nikon-auto-level-sf00001024-2852807323312.html?utm_medium=prd_ads&utm_campaign=22496407165&utm_ad_group=183844197053&utm_content=product&utm_source=google&gclid=CjwKCAiAlMHIBhAcEiwAZhZBUrCqiqKjG0HEYhejuYGvX9GkTS__9Dn_j81QQ8csLuIUgZwbL_d1TRoC58oQAvD_BwE
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7. <https://images.ctfassets.net/1nvkn1423yot/78LHeje1P4D2KuNcnnEHMp/d44edd18de276b0cc3205bbf10c9090a/geo-s5-autolock-productpage-keyfeatures-720x720.jpg?fm=avif&h=950&q=90>

8. <https://www.sisco.com/3-1m-prism-pole>

9. https://www.indiamart.com/proddetail/leveling-staff-4-meter-20970538948.html?utm_medium=prd_ads&utm_campaign=22496407165&utm_ad_group=184992160218&utm_content=product&utm_source=google&gclid=CjwKCAiAlMHIBhAcEiwAZhZBUuPIInYhJx8IWKur_T-DV-3qoxv0_FxNeSAcw14t0VSlm11gqk2jwthoCuE0QAvD_BwE

10. <https://docs.google.com/spreadsheets/d/1B0FUnnrnidgP3zpXM7CYSuqPJYD8nazm/edit?usp=drivesdk&ouid=103124303871165070432&rtpof=true&sd=true>

11. https://docs.google.com/spreadsheets/d/1lwo-__6y0h2wwR0Ur8zHxuXnmXa8Tj0e/edit?usp=drivesdk&ouid=103124303871165070432&rtpof=true&sd=true

12. <https://docs.google.com/spreadsheets/d/10uYP6Pq-hSte0lBrIULnkqYJDglkzMwPt vGIZx6wG5E/edit?usp=drivesdk>

13. Lab Manual of CE331 course.